

Emanuele La Malfa

personal page

I am a postdoctoral researcher at the University of Oxford, where I work on Large Language Models and multi-agent systems. I am also the co-founder and co-director of IDAI, a research institute that focuses on decentralised intelligence as a path towards AI that benefits as many people as possible. My vision centres on **building agents that are natively social** and seamlessly form beliefs, desires, and intentions of other humans and agents. My research and vision on **social training of AI agents** are supported by generous grants from frontier AI labs such as [OpenAI](#), [Google](#), and [Amazon](#), philanthropic organisations such as [Schmidt for Science](#), and the [UK government](#) (ARIA). I have published papers at leading conferences in AI and machine learning, including NeurIPS, ICML, ICLR, AAAI, IJCAI, ACL, and EMNLP.

Work Experience

Postdoctoral Researcher

Jan. 2025 – Current

Rethinking Multi-Agent Systems in the Era of Large Language Models.

University of Oxford

- Principal Investigators: Prof. [Michael Wooldridge](#) and Prof. [Sarit Kraus](#).
- I conduct research on distributed intelligence, coordination and competition of Large Language Models as agents, with a focus on dialogue systems and how to represent their knowledge in a structured, re-usable way.

Co-founder & Co-director

Sept. 2025 – Current

Institute for Decentralized AI (IDAI)

A Project of the Cosmos Institute

- IDAI is a research institute that develops protocols, tools and methods for decentralized intelligence.
- I work on anomaly detection in multi-agent LLM networks and I currently lead several projects on AI safety.
- Grants: [Artificial Intelligence Safety Fund](#) (500,000 USD) and [Responsible AI Amazon Research Awards \(ARA\)](#) (500,000 USD)

Postdoctoral Researcher

Jul. 2023 – Dec. 2025

Benchmarking Large Language Models.

University of Oxford & The Alan Turing Institute

- Principal Investigators: Prof. [Michael Wooldridge](#), [Nigel Shadbolt](#), and [Anthony Cohn](#).
- I conduct research on Large Language Models, with a particular focus on benchmarking their reasoning and planning capabilities. My work is published at the top AI and ML venues (NeurIPS, ICML, ICLR, AAAI, IJCAI, ACL, EMNLP).

Research Assistant

Oct. 2019 – Mar. 2021

Enabling rapid adoption of artificial intelligence through an anonymized data protocol and explainable models.

University of Oxford, UK

- Principal Investigator: Prof. [Marta Kwiatkowska](#).
- Collaboration with [GenieAI](#) and funded through the [InnovateUK scheme](#).

Grants & Fellowships

Build on Trainium: Responsible AI Amazon Research Awards (ARA)

500,000 USD

*Safe, Social Pre-training of LLM Agents. PI: **Emanuele La Malfa**.*

Artificial Intelligence Safety Fund

450,000 USD

*Decentralized Intelligence in the era of LLMs. PIs: **Emanuele La Malfa** and Samuele Marro.*

ARIA: Trust Everything, Everywhere

20,000 USD

*AI Agent Evaluation & Synthetic Content RFP. PIs: **Emanuele La Malfa**, Samuele Marro and Angelo X. Huang.*

Schmidt AI2050 Senior Fellowship Compute

Compute

*Social Training of AI Agents. PI: Prof. Michael Wooldridge and **Emanuele La Malfa**. I wrote the grant proposal.*

Schmidt AI2050 Senior Fellowship

1M USD

Foundations of LLM-based Multi-Agent Systems. PI: Prof. Michael Wooldridge. I wrote part of the grant proposal.

Education

PhD in Computer Science

University of Oxford

Supervisor: Prof. [Marta Kwiatkowska](#). Papers published at AAAI (oral), IJCAI, and EMNLP.

2019 – 2023

Master's Degree

Polytechnic University of Milan

Computer Science and Engineering

2015 – 2017

Bachelor's Degree

Polytechnic University of Milan

Computer Engineering

2011 – 2014

– 2026 –

Benchmarking at the Edge of Comprehension

ICML 2026 (spotlight)

Samuele Marro, Jialin Yu, **Emanuele La Malfa**, Oishi Deb, Jiawei Li, Yibo Yang, Ebey Abraham, Sunando Sengupta, Eric Sommerlade, Michael Wooldridge, Philip Torr

Agentic Business Process Management: A Research Manifesto

Information Systems

Diego Calvanese, Angelo Casciani, Giuseppe De Giacomo, Marlon Dumas, Fabiana Fournier, Timotheus Kampik, **Emanuele La Malfa**, Lior Limonad, Andrea Marrella, Andreas Metzger, Marco Montali, Daniel Amyot, Peter Fettke, Artem Polyvyanny, Stefanie Rinderle-Ma, Sebastian Sardiña, Niek Tax, Barbara Weber

– 2025 –

Large Language Models Miss the Multi-Agent Mark

NeurIPS 2025 (position track, acceptance rate 6%)

Emanuele La Malfa, Gabriele La Malfa, Samuele Marro, Jie M. Zhang, Elizabeth Black, Michael Luck, Philip Torr and Michael Wooldridge

Language Models are Implicitly Continuous

ICLR 2025 (main track)

Samuele Marro, Davide Evangelista, X. Angelo Huang, **Emanuele La Malfa**, Michele Lombardi, Michael Wooldridge

One Language, Many Gaps: Evaluating Dialect Fairness and Robustness of Large Language Models in Reasoning Tasks

ACL 2025 (main track)

Fangru Li, Shaoguang Mao, **Emanuele La Malfa**, Valentin Hofmann, Adrian de Wynter, Jing Yao, Si-Qing Chen, Michael Wooldridge, Furu Wei

When Claims Evolve: Evaluating and Enhancing the Robustness of Embedding Models Against Misinformation Edits

ACL 2025 (Findings)

Jabez Magomere, **Emanuele La Malfa**, Manuel Tonneau, Ashkan Kazemi, Scott Hale

Understanding the Logical Capabilities of Large Language Models via Out-of-Context Representation Learning

EMNLP 2025 (Findings)

Jonathan Shaki, **Emanuele La Malfa**, Michael Wooldridge and Sarit Kraus

– 2024 –

Language-Models-as-a-Service: Overview of a New Paradigm and its Challenges

Journal of Artificial Intelligence Research (JAIR) - **oral presentation at AAAI 2025** - media coverage [here](#) and [here](#)

Emanuele La Malfa, Aleks Petrov, Frieder Simon, Christoph Weinhuber, Raza Nazar, Anthony Cohn, Nigel Shadbolt and Michael Wooldridge

Graph-enhanced Large Language Models in Asynchronous Plan Reasoning

ICML 2024 (main track)

Fangru Lin, **Emanuele La Malfa**, Valentin Hofmann, Elle Michelle Yang, Anthony Cohn and Janet Pierrehumbert

Deep Neural Networks via Complex Network Theory: a Perspective

IJCAI 2024 (main track)

Emanuele La Malfa, Gabriele La Malfa, Giuseppe Nicosia, Vito Latora

A Notion of Complexity for Theory of Mind via Discrete World Models

EMNLP 2024 (Findings)

X. Angelo Huang, **Emanuele La Malfa**, Samuele Marro, Andrea Asperti, Anthony Cohn and Michael Wooldridge

– 2023-2020 –

Language Models Tokenizers Introduce Unfairness Between Languages

NeurIPS 2023 (main track) - [website](#)

Aleksandar Petrov, **Emanuele La Malfa**, Philip Torr, Adel Bibi

The King is Naked: on the Notion of Robustness for Natural Language Processing

AAAI 2022 (main track) – **oral presentation** –

Emanuele La Malfa, Marta Kwiatkowska

On Guaranteed Optimal Robust Explanations for NLP Models

IJCAI 2021 (main track)

Emanuele La Malfa, Rhiannon Michelmore, Agnieszka Zbrzeny, Nicola Paoletti, Marta Kwiatkowska

Assessing Robustness of Text Classification through Maximal Safe Radius Computation

EMNLP 2020 (Findings)

Emanuele La Malfa, Min Wu, Luca Laurenti, Benjie Wang, Anthony Hartshorn, Marta Kwiatkowska

Tacit Coordination of Large Language Models

Under review

Ido Aharon, **Emanuele La Malfa**, Michael Wooldridge, Sarit Kraus**An End-to-end Planning Framework with Agentic LLMs and PDDL**

Under review

Emanuele La Malfa, Ping Zhu, Samuele Marro, Sara Bernardini, Michael Wooldridge**Fixed Point Explainability**

Under review

Emanuele La Malfa, Jon Vaddillo, Marco Molinari and Michael Wooldridge**A Scalable Communication Protocol for Networks of Large Language Models**

Under review

Samuele Marro, **Emanuele La Malfa**, Jesse Wright, Guohao Li, Nigel Shadbolt, Michael Wooldridge, Philip Torr

Conferences and Workshops Organization

Benchmarking Large Language Models

28/11/2023

Workshop Organizer

The Alan Turing Institute, London, UK

LOD2020, LOD2021, LOD2022, LOD2023, LOD2025

General Chair (2025), Conference Chair (others)

Lake District/Siena/Grosseto

Invited Lectures, Talks, and Presentations

Code Simulation Challenges for Large Language Models

02/02/24

Group Talk

Bocconi, Italy

On Robustness for Natural Language Processing

19/04/2023

Group Talk

ICREA, Barcelona

On the Notion of Robustness for Natural Language Processing

17/01/2023

Departmental Talk

King's College University of London, UK

Robustness for Natural Language Processing

22/04/2022

Lecture – *Deep Fridays*

University of Bologna, Italy

Explainable AI

04/03/2022

Lecture – Advanced Artificial Intelligence Course

Royal Holloway University of London, UK

Academic Reviewing and Volunteering

The Alan Turing Institute - Reviewer

October - December 2023

Reviewers for the Turing Fellow Program - Panel "Fundamental Research in Data Science and AI"

Ukrainian Global University - Interviewer

April-June 2023

I interviewed Ukrainian students who want to study in a partner university abroad.

The Kharkiv and Przemyśl Project

August 2022

I spent a week in Przemyśl (Poland) as a volunteer to help refugees who arrived (returned) from (to) Ukraine.

Ukrainian Global University - Interviewer

June-May 2022

I interviewed a dozen of prospective undergraduate Ukrainian students who want to study in a partner university abroad.

Eutanasia Legale - Volunteer

July 2021

I have collected signatures for a referendum to decriminalize euthanasia. The overall campaign gathered 1.2 million valid signatures.

Academic Service

Conference Reviewer: ICML, NeurIPS, ICLR, ACL, EMNLP.

Teaching Experience

Deep Learning in Healthcare

2024 (HT)

Practical sessions

University of Oxford, UK

Machine Learning

2023 (MT)

Classes

University of Oxford, UK

Probabilistic Model Checking

2023 (MT)

Practical sessions

University of Oxford, UK

Ethical Computing in Practice

2023 (HT)

Practical sessions

University of Oxford, UK

Deep Learning in Healthcare

2023 (HT)

Practical sessions

University of Oxford, UK

Probabilistic Model Checking

2022 (MT)

Practical sessions

University of Oxford, UK

Machine Learning

Classes

Probabilistic Model Checking

Practical sessions

Fundamentals of Computer Science

Practical sessions

2021 (MT)

University of Oxford, UK

2021 (MT)

University of Oxford, UK

2016 (Oct.-Dec.)

Polytechnic University of Milan, Italy

Mentoring and Student Supervision

Samuele Marro

I co-supervised his [Master's thesis](#), published at ICLR'25. Samuele is a PhD student at the Dept. of Engineering, University of Oxford.

Angelo Huang

I co-supervised his [Bachelor's thesis](#), published at EMNLP'24. Angelo is a Master's student in Computer Science at ETH.

Ping Zhu

I supervised his Master's thesis at Oxford. Ping is doing an MSc in advanced computer science at the University of Oxford.

Alberto Zurini

I co-supervised, with Alberto Cazzaniga, his Master's thesis. Alberto is a Master's student in Computer Science at the University of Udine.

Giovanni Monea

PhD student in Computer Science at Cornell University.

Lead the Future

Simone Alghisi

PhD student in Information Engineering and Computer Science at the University of Trento.

Lead the Future

Andrea Cerutti

Master's student in Computer Science and Engineering at the Polytechnic University of Milan.

Lead the Future

Riccardo Inghilleri

Master's student in Computer Science at the Polytechnic University of Milan.

Lead the Future

Orazio Torre

Bachelor's student in Computer Science at the University of Salerno.

Lead the Future

Annalaura Pegoraro

Bachelor's student in math at the University of Padova.

Lead the Future

Mattea Busato

Bachelor's student at Bocconi University.

Lead the Future

Alessandro Soccol

Master's student at the University of Cagliari.

Lead the Future

Nathan Gebreamlak

I supervised Nathan for a term for the Williams-Exeter exchange program (2025) - Nathan is a bachelor's student at Williams College (US).

Charlie Tharas

I supervised Charlie for a term for the Williams-Exeter exchange program (2025) - Charlie is a bachelor's student at Williams College (US).

Saad Waheed

I supervised Saad for two terms for the Williams-Exeter exchange program (2024) - Saad is a bachelor's student at Williams College (US).

Alisa Kanganis

I supervised Alisa for two terms for the Williams-Exeter exchange program (2024) - Alisa is a bachelor's student at Williams College (US).

Simon Socolow

I supervised Simon for a term for the Williams-Exeter exchange program (2024) - Simon is a bachelor's student at Williams College (US).

Tutoring and Mentoring

Williams-Exeter Exchange Programme

2023-2025

Tutoring Saad Waheed, Alisa Kanganis, and Simon Socolow (Williams-Exeter Programme exchange students in machine learning).

University of Oxford

2022

Tutoring Edward Kusel and Aleksandar Radoslavov for their part-B projects (undergraduate in Computer Science).

Lead the Future - Mentor

2022-current

Lead the Future helps Italian STEM talents find their path to brilliant careers. I am currently mentoring 9 students.

Other Skills

Programming Languages

Python, C++ (CUDA). In the past, I used web languages (e.g., JavaScript) and low-level languages (C and x86 Assembly).

Libraries

PyTorch, Numpy, CUDA.